

Figure 4: UHCS microstructure samples showing 5 microconstituent classes.

radiographic (X-ray) images of weld joints with 4 classes: lack of penetration, porosity, cracks, and no defect. We use 24 training images (6 per class) and 240 test images (60 per class, randomly sampled from a larger test set).

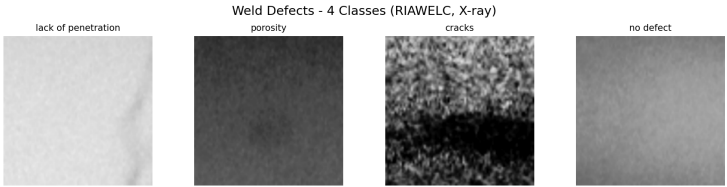


Figure 5: Weld radiograph samples showing 4 defect classes (RIAWELC dataset).

4.2 Baselines

For each task, we compare against four baselines:

- Base model zero-shot (Base ZS).** Qwen2.5-VL-3B-Instruct [14] evaluated with the classification prompt and test image only, without any fine-tuning or reference examples.
- Frontier model few-shot (GPT-4.1 FS).** GPT-4.1 evaluated with the classification prompt plus a grid of reference images (one per class) provided alongside the test image. The trained models (Direct and CoT-augmented) do not receive reference images at inference time — their domain knowledge is learned during fine-tuning rather than provided at evaluation.
- Direct LoRA.** The small model fine-tuned on N image-JSON pairs (Approach A). This isolates the contribution of CoT distillation by providing a baseline with the same images, same model, and same hyperparameters but without reasoning augmentation.
- Answer-conditioned CoT LoRA.** The small model fine-tuned on $4N$ reasoning-augmented pairs (Approach B). This is our proposed method.

4.3 Training Configuration

All training uses the configuration described in Table 1. Training images are selected randomly (seed 42) from the training split. The same seed ensures both Direct and augmented approaches train on the same images. The GPT-4.1 API (accessed via Azure AI Foundry) generates CoT descriptions with temperature 0.7 and maximum 512 tokens.

All models reported in this paper were trained on $2 \times$ NVIDIA V100 16GB GPUs (Azure ML), with the model split across GPUs using memory mapping. To estimate single-GPU